

Cover Letter for PLOS ONE

Dear Editors of PLOS ONE,

Please consider my manuscript, “Computational surveillance of Colombian public pharmaceutical procurement using public administrative data: a reproducible analysis of a closed 2020-2025 cohort,” for publication in PLOS ONE.

This study describes and evaluates a reproducible computational infrastructure for surveillance of Colombian public pharmaceutical procurement using public administrative data from SECOP-II, INVIMA, and SISMED. The manuscript reports the integration of these sources into an open analytical workflow combining relational storage, statistical anomaly detection, and a public-facing observatory. Using a closed analytical cohort of 162,271 pharmaceutical contracts from 2020 to 2025, I identified statistically atypical procurement patterns, documented marked market concentration, and generated auditable outputs from otherwise fragmented public data sources.

I believe the manuscript is well suited to PLOS ONE because its principal contribution is methodological and reproducible rather than journal- or country-specific. The work presents a transparent computational framework that can be evaluated, reused, and adapted in other settings where public procurement data are available but underused for systematic surveillance. In addition to its relevance for pharmaceutical policy and public-sector auditing, the study contributes to the broader literature on open computational infrastructures, administrative data reuse, and applied analytical methods for public-interest oversight.

The manuscript uses exclusively public, non-personal administrative data. No clinical, laboratory, or identifiable patient-level data were used. Under Colombian regulations, the study is classified as research without risk. The blockchain-based payment-cycle component is presented only as a complementary functional proof of concept and not as the central empirical contribution of the manuscript; it does not report real-world operational timings.

This submission is original, has not been published previously, and is not under consideration elsewhere. I have reviewed the manuscript, approve its submission, and agree to be accountable for the work. I have no competing interests to declare. The study was conducted without external funding.

For reproducibility, the article-specific package is archived at Zenodo under DOI <https://doi.org/10.5281/zenodo.19074137> and is also available in the dedicated public repository at <https://github.com/zswamtech/BigLoI-PLOS-ONE-paper>. The public observatory derived from the project is not cited as a live URL in this submission version because its production deployment is being finalized; the archived package therefore serves as the authoritative record for editorial evaluation.

Thank you for your consideration.

Sincerely,

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